Phase 3E | Clearance resolution and prototype preparation

PULSIPHER FARMS / MOAPA VALLEY CORN MAZE

Make the next physical work measurable

PF-TRAIN-S3E-R0 | 18 October 2026 UTC

Preliminary engineering development - not for vehicle fabrication, construction, procurement release or passenger operation. The ground-level layout mockup instructions are for checking space and movement. Vehicle parts, elevated occupied structures and powered tests require their separate engineering releases.

This phase follows the wheel-clearance conflict identified in Phase3D. It compares coordinated frame changes, provides a full-size interior trial layout and defines exactly what suppliers and the first prototype must demonstrate. No supplier response, occupied trial, physical test or professional approval is assumed.

New work	Purpose
Clearance alternatives	Compare the retained low floor, a raised-floor study and a local frame/seat-support revision without cutting notches into main girders.
Actual-size mockups	Lay out regular and accessible spaces at ground level. Use rated seating and movable non-load-bearing markers; record actual adult/device circulation before freezing the interior.
Supplier verification	Provide component-specific technical packets and evidence fields, with responsibilities and dependencies across the complete vehicle.
First-article preparation	Define measurement channels, test stages, hold points and release records before hardware procurement and testing.
Coordinated files	Readable drawings, full-size CAD coordinates, rotating frame/wheel studies, editable calculations and blank observation records.

01 | Controlled continuation of Phase 3D

The current saved Phase3D component-integration report was read on 18 October 2026. Its last-modified time was 05:40:38 UTC that day; the public technical site was confirmed at version6. Earlier reports, drawings and models remain available. Phase3E is a new, independently identified revision of the studies, not a construction release.

Treatment	Design basis
Retained owner directions	0% longitudinal grade target; 36-inch evaluation gauge; diesel locomotive; three regular coaches and one separate accessible coach; enclosed heating/cooling; hydraulic service-brake preference. North Pole location remains deferred; passengers disembark there.
Retained cabin and train	306 x104 in clear cabin; 312 x110 in body; 400 in frame; bogies x45/261 in; 32 regular places, 11 accessible fixed places and two wheelchair places. Occupancy remains subject to trials and approval.
Retained calculation loads	7,040/7,535 kg tare allocations; 54,755 kg nominal and 59,840 kg stress train; 13,200/11,000 kg upper individual-car cases and 65,100 kg upper train. These are calculation inputs, not equipment ratings.
Changed only in named alternatives	Floor elevation and selected crossmember/seat-support geometry are varied explicitly. The 28 in baseline is not silently replaced by a raised floor.
Still missing	Actual configured bogie/wheel/brake/coupler drawings, supplier acceptance, occupied mockup observations, surveyed alignment and accepted design/fabrication/test criteria.

The public technical edition excludes business forecasts and attraction scenes. The owner and farm crew can progress the ground-level layout and documented information gathering now. Vehicle fabrication, load tests, track construction and passenger use retain their engineering gates.

02 | Use this issue in a decision sequence

Recommended next review: investigate the E1 raised-floor architecture first because it retains the stronger crossmember orientation and existing seat-support geometry. Its 32-inch floor is still a trial dimension: the larger movement-envelope sensitivity remains unresolved. Keep E2 as a low-floor comparison; its small nominal gap, weaker crossmembers and possible seat-anchor uplift require additional design. Select the final height only with the configured bogie, boarding and stability evidence.

Decision	Evidence and next action
Which frame/floor direction?	Review all three alternatives with the complete configured bogie envelope. A positive cylinder-proxy gap does not establish flange, suspension, brake or full-motion clearance.
Does the interior work?	Build the ground-level layout, use six adults for a repeatable regular-coach trial, and check representative mobility devices and companion movement in the accessible layout. Record measurements, limitations and proposed changes.
Which supplier configuration?	Return one coordinated drawing and rating set for the exact offered assembly. Resolve scope gaps and deviations before committing to parts.
What can be fabricated?	Release one controlled first-article package after structural, wheel-rail, braking, electrical and boarding interfaces are accepted. Freeze the approved bill of materials and inspection/test limits.
When can it be duplicated?	Close measured mass, workmanship and staged test evidence. Resolve discrepancies on the first article before copying it into the other coaches.

Open fields remain blank in the evidence forms. A prepared form is not a completed check. Numerical sensitivities describe stated assumptions and must not be converted into supplier ratings, code acceptance or test limits without the responsible review.

03 | Frame alternatives and load-path consequences

PF-TRAIN-S3E-R0 · 18 October 2026 UTC · **PRELIMINARY ENGINEERING STUDY - NOT FOR FABRICATION OR CONSTRUCTION**

Phase 3D found that a nominal 596 mm wheel could occupy the same space as a four-inch-deep crossmember beneath the proposed 28-inch floor. Phase 3E compares two coordinated responses: raise the existing frame, or retain the low floor with shallower members in the wheel zones and move the inner seat-support rails. **Neither response establishes fit for a real bogie.** The latter has little nominal vertical margin and introduces weaker bending orientation and more demanding seat connections.

Keep the owner's 0% longitudinal grade target, 36-inch evaluation gauge, 104-inch clear cabin width, eight two-person bench rows per side and 32 regular-coach places. The retained 7,040 kg tare allocation and 13,200 kg upper structural study case do not change. This study modifies support geometry beneath the interior; it does not narrow the aisle or remove seats.

1. Three complete, comparable arrangements

Option	Finished floor	Structural arrangement	Bare modeled steel
E0	28 in	Retained D2; reference with identified interference	1,527.29 kg
E1	32 in	Same complete D2 frame raised four inches	1,527.29 kg
E2	28 in	Four shallow central members; inner seat rails shifted outward	1,525.44 kg

Each option contains 110 disjoint member envelopes. E0 preserves the predecessor for traceability. E1 is a comparison, not a decision to raise the floor. E2 is the drawing/model focus for exploring a low-floor response, not a fabrication selection. Canonical coordinates, section orientations, support surfaces and panel bounds are provided in structure_e.json; the portable script and input file reproduce both geometry and calculations.

E2's 1.84 kg reduction is only the consequence of shorter seat-rail segments at wider transverse nodes. It is not an optimization benefit large enough to justify the new interfaces. All three remain above the former 1,500 kg frame allocation before connections, mounts, draft structure, stairs, guards, coatings and other excluded parts.

The practical next decision is whether a documented bogie can support an acceptable raised floor. E1 preserves the existing member orientation and seat base and therefore provides the simpler structural arrangement to investigate first. If a low floor is essential, E2 requires a complete wheel-envelope, seat-base and joint review; its small nominal gap should not drive procurement.

2. What physically changes in E2

All options retain two continuous HSS 12×4×¼ main girders at y±34, 12-inch-deep bolster segments at x45/261, and the 400-inch frame from x−47 to 353. Coordinates are inches: x0 is the front inside wall, y0 the vehicle centreline, and z0 top of rail. No main girder or bolster is notched.

The cabin crossmember stations remain −3, 21, 45, 69, 93, 117, 141, 165, 189, 213, 237, 261, 285, 309. At **x21, 69, 237 and 285**, only the central transverse segment from y−32 to 32 changes orientation. Its existing HSS 4×2×⅛ is rotated so that it is **two inches vertically and four inches longitudinally**. The outer transverse arms remain four inches deep and two inches wide. Other ordinary crossmembers remain unchanged.

This is an orientation change of a complete catalog tube, not a proposal to cut away its bottom wall. The central member still needs to transfer forces into the main-girder walls. Its neutral-axis height, attachment

footprint and joint stiffness differ from the adjoining members; these are engineering interfaces, not details solved by drawing touching rectangles.

The original three-inch-deep inner seat rails at y±20 also pass through the wheel area. E2 moves these lines to **y±26**, while retaining the outer lines at y±44. The two-person benches, row pitch and cabin circulation remain unchanged; their actual bases have not been selected. The new foot lines are a supplier-facing geometry proposal and must be checked against the complete wheel envelope.

The four-inch-wide rotated nodes shorten adjacent inner rail segments by one inch at each affected end. Four nodes, two inner rail lines and two approaching segments remove 16 inches of HSS 3×1×⅛ in total. The script accounts for this explicitly. Outer rail segments retain their original lengths because the outer transverse arms retain their two-inch width. The three-inch seat rails also extend one inch below the new two-inch central tube at their ends, so a developed joint cannot be assumed from the visible face overlap.

3. Rotated sections use the correct published bending properties

Atlas's manufacturer-hosted table was checked on 18 October 2026 C3ES01.

Orientation of HSS 4×2×⅛	Vertical depth	Longitudinal width	I used	S used
Upright	4 in	2 in	2.65 in⁴	1.33 in³
Rotated	2 in	4 in	0.898 in⁴	0.898 in³

Both orientations retain the same nominal 4.75 lb/ft mass and 0.116-inch design wall. The rotated section therefore has only 33.9% of the original vertical bending inertia. The model encodes the rotated section as a distinct C4L_R entry; it does not apply strong-axis properties to a shallow shape. Visual meshes show nominal sharp-corner tubes; published properties and weights govern calculations, not approximate mesh volumes.

No allowable stress, material grade, toughness, fatigue category or weld size is inferred from this table. Exact stock and lead time remain unverified. Local tube-wall and connection behavior must be assessed separately from member bending C3ES02.

4. Seat loads change even though the passengers do not

Each study bench retains two 145 kg adults plus 15 kg assumed bench mass: 305 kg total. Its resultant transverse CG is assumed at y32, the centre of the forty-inch bench. For the two foot lines, equilibrium gives:

near-line share=(outer−CG)/(outer−inner); outer-line share=(CG−inner)/(outer−inner).

Each line has two fore/aft feet at row-centre±6 inches. E0/E1's y20/44 lines each carry half the bench load, or 747.76 N per foot. E2's y26/44 lines carry two-thirds and one-third respectively: **997.01 N per near foot and 498.50 N per outer foot**. Equal quarter-load feet would be inconsistent with this assumed seat CG.

The calculation transfers every foot through its longitudinal rail to adjacent transverse nodes, then through the transverse beams to the mains. Physical rail segments end at member faces; the analytical supports are at node centrelines. Joint offsets and actual anchor stiffness require separate analysis.

Static seated-load result	E0 / E1	E2
Maximum seat-rail bending stress	12.76 MPa	17.01 MPa
Maximum seat-rail elastic deflection	0.0713 mm	0.0951 mm
Largest seat-derived crossmember-node load	4.736 kN	4.736 kN
Maximum crossmember deflection from these seats	0.163 mm	0.571 mm

A separate asymmetric sensitivity places one 145 kg adult at y22, representing the inner half of the bench, with the 15 kg bench resultant still at y32. Their combined CG is y22.9375, inward of E2’s proposed y26 support. E2 then requires **−267 N on the outer support line, or −133.5 N at each outer foot**, while E0

retains positive reactions. These are assumed positions, not an anchorage standard, but they demonstrate the need for positive seat retention and a qualified bench frame with its longer inner overhang.

The unchanged total node load does not mean unchanged connections. The nearer foot load grows 33.3%, and transverse flexibility increases. Reinforced anchor details must transfer forces into steel without crushing the floor finish, bedding or honeycomb. Actual seat mass, CG and supplier anchorage requirements replace these study assumptions before release.

5. The floor remains supported, but its response needs coordination

The original transverse panel seams at x45, 93, 141, 189, 237, 285 remain on transverse members. Stock-layout longitudinal seams remain over the main girders at y±34. There are seven full-width panel rectangles in the model; their dimensions are supported-layout geometry, not cutting or installation instructions.

Maximum centre spacing remains 24 inches. In E2 the central clear gaps adjacent to a rotated tube are 20-21 inches depending on the neighboring member; ordinary gaps remain 22 inches. Outer floor areas still see the original member widths. At seam x237, nominal central support width becomes four inches while the outer support remains two inches. A continuous panel seam therefore does not have one uniform bearing detail across the car.

The elastic transverse model retains four-inch-deep outer arms and uses the rotated section only between the main-girder support centrelines. It integrates moment over the actual strong/weak stiffness zones. Connections are idealized as continuous; their real flexibility and local wall behavior are absent.

Alternative uniform-floor sensitivity, 24-inch tributary	Upright centre	Rotated centre
4.0 kPa: maximum member stress	27.28 MPa	40.41 MPa
4.0 kPa: maximum displacement	0.749 mm	2.210 mm
4.8 kPa: maximum member stress	32.74 MPa	48.49 MPa
4.8 kPa: maximum displacement	0.899 mm	2.652 mm

These pressures are analyst comparison cases, not adopted code loads or extra payload added to the seated-load case. The results are demands, not passes against an invented allowable. Panel-only deflection on immovable supports cannot establish finished-floor performance when the support frame itself deflects. The engineer must coordinate panel bending/shear, support movement, seams, caster/heel loads, vibration and accepted serviceability limits.

6. Nominal vertical space is small in E2

Keep the 25.30 mm trial stack with a 0.75-inch panel. At a 28-inch finished floor, steel is at 685.9 mm. An upright four-inch crossmember ends at 584.3 mm; the rotated two-inch member ends at 635.1 mm. A nominal 596 mm wheel with 25.4 mm relative upward motion reaches 621.4 mm before additional flange/profile or roll effects. Simple plane arithmetic therefore leaves **only 13.7 mm** under the shallow member.

That number is not the actual design clearance. The companion all-member sweep checks the wheel proxy against the complete frame, including seat rails, and separates nominal geometry from additional envelope sensitivities. Actual wheel profile, flange/rim extent, suspension motion, roll, wear, tolerance and required reserve remain supplier/engineer inputs. If a one-inch floor panel is adopted with finished height held fixed, steel lowers 6.35 mm and this simple gap falls to 7.35 mm. No thicker panel is silently substituted.

E1 raises every frame member 101.6 mm. Its retained 12-inch bolster underside rises from 381.1 to 482.7 mm, but a real support top, attachment stack and relative-motion clearance must still fit below it. Raising the body does not prove that a compatible bogie exists.

7. Global mass, CG and release decisions

All alternatives use the same 13,200 kg vertical comparison: 4,880 kg of occupied benches, 145 kg centred crew and 8,175 kg residual allocation spread over 400 inches. Body stiffness is not credited. The seat changes preserve total vertical force and transverse moment, so the main-girder static response remains 56.53 MPa maximum bending stress and 5.035 mm maximum deflection in this particular case. This does not supersede asymmetric crowd, braking, torsion, track twist, fatigue or jacking/lifting cases.

E1 also raises the CG of whichever masses move with the frame. The correct relationship is whole-vehicle CG increase=(raised mass/total mass)×101.6 mm. Raised-mass fractions of 0.50/0.75/1.00 give 50.8/76.2/101.6 mm increases. Actual raised mass and sprung/unsprung distribution are unknown; these are sensitivities, not validated stability margins. Revisit roll/wheel unloading, braking transfer, roof clearance, stairs, accessible boarding and station interfaces together.

Before structural cutting, obtain controlled bogie and wheel drawings, an agreed movement envelope and clearance reserve, a complete floor construction and actual seat-base requirements. The responsible engineer can then choose the architecture and release local walls, joints, materials, tolerances and inspection criteria. Farm crews can assemble a ground-level layout mockup with rated seating to check the retained cabin and proposed seat-foot positions; they should not build shallow structural substitutions from these studies. The calculations make the consequences reviewable while preserving the unresolved decisions explicitly.

04 | Wheel-clearance envelopes and required returns

PF-TRAIN-S3E-R0 · 18 October 2026 · **PRELIMINARY ENGINEERING STUDY - NOT FOR FABRICATION**

The 28-inch floor can clear the previously studied wheel cylinder if the frame is changed, but that does not yet demonstrate a compatible coach. Phase3E compares two concrete repairs to the retained D2 frame and checks **all110 modeled members**, including seat rails omitted from the earlier local wheel/crossmember calculation. The result is a useful design choice: a raised floor offers more clearance, while a low-floor bridge arrangement preserves boarding height but leaves much less room for the actual wheel profile and suspension motion. Neither alternative is selected for construction.

1. Retained basis and the meaning of the wheel model

E0 retains the Phase3D D2 arrangement, 28-inch finished floor, 25.30 mm floor stack,12-inch mains/bolsters and4-inch ordinary crossmembers. It retains32 regular seats, the104-inch clear cabin and0% longitudinal grade target. E1 raises the same frame and finished floor4 inches, to32 inches. E2 keeps28 inches, rotates only the four central crossmember segments atx21/69/237/285 to2-inch depth and4-inch longitudinal width, and moves the inner seat-attachment lines fromy±20 toy±26 inches. Outer crossmember arms, main girders and central bolsters remain unchanged. Member geometry comes directly from the structural module; no independent clearance drawing silently changes the frame.

The **596 mm rolling diameter and1165 mm wheelbase** are documented Railteco RTKZ1 reference dimensions C3EC01 C3EC02, rechecked2026-10-18 UTC. That product remains rejected as supplied: its762 mm gauge differs, and Phase3D identified a load mismatch. The101.6 mm cylinder width and transverse centre planes at±457.2 mm are **analyst proxies**, not manufacturer dimensions. Centre-plane spacing is not wheel back-to-back or proof of36-inch track compatibility. No real flange, axlebox, bearing, brake or bogie support shape has been invented.

The diameter/wheelbase sweep adds400/500 mm diameters and1000/1500 mm wheelbases as sensitivities. Changing a wheel’s diameter can alter contact stress, bearing/axle design, brake torque and thermal capacity. A smaller cylinder fitting the drawing is not a component selection. GHH-BONATRANS explicitly develops wheelsets around vehicle, bogie, brake and track requirements C3EC03; a controlled project-specific drawing must replace the proxy before release.

2. Exact nominal-cylinder checks now include the seat rails

For zero relative roll, the script rotates each member's horizontal footprint into bogie coordinates, clips it to the wheel-width strip, and finds the smallest distanceδ from that footprint to the axle's longitudinal coordinate. The maximum cylinder height over the footprint is:

zmax = rrolling + √[(rrolling + radial addition)² −δ²] + relative heave.

Subtract it from the member underside to obtain the geometric gap. The calculation also checks horizontal overlap and vertical intersection. This is exact for the stated cylinder and rectangular member envelope; it does not represent an actual wheel profile. Five yaw states are checked, at0,±half and±full of the rigid-circle yaw for the specified wheelbase and30 m radius. For1165 mm, the extreme is±5.24745°. These are finite packaging states, not a continuous suspension or reverse-curve certification.

Configuration	Rest cylinder, no heave	Same cylinder with25.4 mm relative upward motion	Governing modeled member
E0: retained28-inch floor	−11.70 mm	−37.10 mm	Central crossmember atx21; equivalent conflicts occur at69/237/285
E1: entire frame raised4 inches	+89.90 mm	+64.50 mm	Same crossmembers
E2: four shallow bridges and moved inner seat rails	+39.10 mm	+13.70 mm	Shallow central crossmembers

Negative values are confirmed intersections of these analytical solids. Positive values are geometric gaps at checked states, not accepted operating clearances. The heave allowance is a trial, not a measured spring travel or required code value.

The new full-member check finds an additional E0 issue: several3-inch-deep inner seat rails aty±20 also intersect the25.4 mm heave cylinder, by up to **11.70 mm**. Repairing only the four transverse members would leave that conflict. E2's moved inner rails lie outside the nominal proxy's horizontal sweep. This move requires a new seat base and altered reactions; it is coordinated with the structural analysis, not an instruction to relocate an existing seat's bolts. The companion single-inner-adult sensitivity also produces133.48 N static uplift at each E2 outer foot, requiring a designed retention path; ordinary equal-load seat feet cannot be assumed. Neither arrangement adds seat capacity or intrudes on the retained aisle.

The full nominal sweep contains81 cases: three configurations, three diameters, three wheelbases and three heave values. For596 mm wheels and25.4 mm heave:

Wheelbase trial	E0 minimum gap	E1 minimum gap	E2 minimum gap
1000 mm	−34.40 mm	+67.20 mm	+14.05 mm
1165 mm	−37.10 mm	+64.50 mm	+13.70 mm
1500 mm	−29.78 mm	+71.82 mm	+16.42 mm

Wheelbase changes where the circle meets the members; it does not simply increase clearance. E2's wider rotated member also occupies more longitudinal space, which the calculation includes. The source records identify the actual limiting member, wheel, yaw and gap for every case.

3. Radial profile and roll can consume the apparent repair

A wheel's maximum radial outline can extend beyond the rolling-radius cylinder. Until the manufacturer returns a real profile, the study adds0/25/35 mm to the outer cylinder radius across its full width. The added values are **unselected dimensional allowances**, not alleged RTKZ1 flange heights. This inflated cylinder is deliberately simple; it still cannot represent an unknown brake or axlebox.

For relative roll, the model first applies bogie yaw, then rotates the wheel proxy about a longitudinal reference axis atz381.1 mm and applies vertical relative heave. The frame remains fixed. This is a mathematical relative-motion transform, so the transformed wheel need not remain on the nominal rail plane; it is not a claim that the real wheel leaves the track. The actual roll centre and correlated suspension motions remain unknown.

The exact bounding box of this transformed cylinder is tested against every member. Its half-extent along coordinatei is router$\sqrt{(1-a_i^2)}+(\text{width}/2)|a_i|$, whereai is the cylinder-axis direction component. A positive gap with overlapping horizontal boxes proves separation for that bounded proxy at that sampled state. A negative bounding-box gap identifies **possible interference**, which needs detailed intersection analysis; it does not prove that the real solids collide.

Combined trial	E0 bound gap	E1 bound gap	E2 bound gap
P2:25 mm radial addition,25.4 mm heave,±2° roll	−81.47 mm	+20.13 mm	−30.67 mm
P3:35 mm radial addition,50.8 mm heave,±3° roll	−126.42 mm	−24.82 mm	−75.62 mm

Yaw uses the same five states; roll checks both extremes and zero. E2's13.70 mm nominal gap therefore cannot be treated as proof that the bridge repair is sufficient. E1 offers substantially more space, but even its32-inch floor does not clear the widerP3 bound. Conversely, a box overlap is not evidence that32 inches definitely fails the eventual OEM arrangement. The engineer must replace the allowances with actual correlated motions and profiles, then evaluate their required combinations continuously or with an accepted bounding method.

4. How much elevation would the wheel proxy require?

For a uniform vertical shift of the same geometry, new floor = present floor +(comparison reserve − minimum gap)/25.4, in inches. This is a **wheel-only** result: the bogie support stack, braking equipment, boarding, centre of gravity and all omitted parts remain separate constraints. A10 mm or25.4 mm reserve below is merely a comparison target; neither is an adopted minimum clearance.

Geometry family and trial	Floor for zero proxy gap	Floor for10 mm comparison reserve	Floor for25.4 mm reserve
E0/E1 crossmember geometry, P1 nominal/heave	29.461 in	29.854 in	30.461 in
E0/E1 geometry, P2 bound	31.207 in	31.601 in	32.207 in
E0/E1 geometry, P3 bound	32.977 in	33.371 in	33.977 in
E2 shallow-bridge geometry, P2 bound	29.207 in	29.601 in	30.207 in
E2 shallow-bridge geometry, P3 bound	30.977 in	31.371 in	31.977 in

P2/P3 heights are sufficient heights for the sampled bounding-box screen, not exact minimum heights for the rotated cylinder or actual running gear. These values quantify the trade, not a new floor selection. For example, a32-inch E1 floor clears theP2 bound by20.13 mm, but it does not provide the illustrative25.4 mm

reserve. A bridge arrangement could recover about2 inches relative to that same proxy, while demanding new joints, seat bases and structural checks. The independent central-bolster inequality remains: actual support height plus attachment and required movement/tolerance allowance must fit below the12-inch bolster. A shallow ordinary crossmember does not raise that support plane.

5. Why simply moving the offending crossmembers is not released

A diagnostic relocation moves21→9,69→81,237→225 and285→297 inches, away from the axle crown region. It is **not a canonical E alternative**, cut list or approved repair. With fixed bolsters at45/261, this creates36-inch floor bays instead of24-inch bays and moves panel seams and seat-rail support nodes. At equal uniform load and equal span-to-deflection criterion, the bending-only requiredEI increases by $(36/24)^3 = 3.375$. Panel shear, rolling/point loads, edge support, joints and seat loads need further checks.

Moving a steel member out of a wheel’s way while leaving its panel seam or seat load unsupported does not produce a complete floor. The coordinated E2 bridge study is more useful because it retains the node stations and gives a defined load route. Its local connection and fatigue design still require engineering; no tube may be rotated, notched or welded solely from this packaging study.

6. Supplier return and go/no-go sequence

The editable return map identifies eight required interface packages. Obtain controlled wheel profiles and new/worn diameters; transverse wheel setting and matched rail/turnout geometry; actual wheelbase and correlated yaw/roll/pitch/heave/lateral travel; support and sidebearer heights; complete running-gear/brake CAD; engineer-set clearance combinations; E2 seat/floor/joint details if pursued; and first-article validation procedures. Every actual value remains blank until supported by evidence.

Advance a configuration only when three tests agree. First, the component and wheel/rail system must meet the load, fatigue and application requirements; a dimensional fit cannot cure the rejected catalog assembly’s gauge or axle-load mismatch. Second, detailed loaded/unloaded and new/worn geometry must clear the complete frame, body, stairs, couplers, brakes and infrastructure under accepted combinations. Third, as-built measurements and controlled first-article tests must corroborate the drawings and calculation margins.

If the nominal proxy already intersects, revise geometry before a fit release. If only the bounding box overlaps, keep the decision open and obtain the detailed model; do not classify the product as failed or clear from that bound alone. If every checked proxy clears but an OEM field or accepted criterion is missing, the result remains unresolved. This sequence allows useful supplier and fabrication review now without representing the study as permission to order running gear or carry passengers.

05 | Full-size coach mockups and accessible interfaces

PF-TRAIN-S3E-R0 · 18 October 2026 UTC · **Ground-level layout study only. No vehicle fabrication, elevated passenger loading or accessibility approval. All trial records are blank.**

Build this inexpensive mockup on an existing firm, level floor before fixing the seating or doors. It preserves the 306 × 104 in clear cabin and tests how much of that space remains usable with adults, mobility devices and companions. Use six suitable commercial chairs for occupied observations. Tape and lightweight cards represent the vehicle; they carry no people. Farm crew can prepare this layout without constructing the provisional steel frame.

Set out the exact floor plan

Use inches throughout. The front inside wall is x = 0; the coach centreline is y = 0. Positive x runs toward the rear. The actual existing floor is z = 0, independent of the vehicle's rail datum. Reserve an unobstructed **520 × 302 in work area (43 ft 4 in × 25 ft 2 in)** for both side approaches and end movement. This is a mockup work allowance, not a station clearance requirement.

Item	x limits	y limits / dimensions
Clear cabin	0-306	−52 to +52
Outer body reference	−3-309	−55 to +55
Front / rear platform outlines	−47 to −3 / 309-353	−55 to +55
Regular centre aisle	0-306	−12 to +12; 24 clear
Regular seat starts, each side	31, 63, 95, 127, 159, 191, 223, 255	Left −52 to −12; right +12 to +52
Each regular bench plan	20 long	40 wide; two nominal places
Regular end doors	−3-0 and 306-309	−16 to +16; 32 clear
Accessible route	60-306	−12 to +24; 36 clear
Wheelchair bays WC1 / WC2	60-120 / 120-180	−52 to −16; each 60 × 36
Accessible companion seats	80-100 / 140-160	+24 to +48
Accessible side doors	9-51	At both side walls; 42 clear
Accessible end doors	−3-0 and 306-309	−12 to +24; 36 clear
Front turning circle	Centre x = 30	Centre y = 6; diameter 60
Flat bridge reservations	12-48	+55 to +91 and −91 to −55
Station turning reservations	0-60	+91 to +151 and −151 to −91

The accessible coach also retains three rear rows beginning x = 190, 222 and 254. Each has a 20 × 40 in two-place left bench and a 20 × 28 in right seat; together with the companions this gives 11 fixed places and two wheelchair bays geometrically. Seating and admission capacities remain unresolved.

Tape clear boundaries at their inner edges. Check the cabin's two diagonals: **323.1903 in**. Use ±1/8 in as the selected layout goal, not a fabrication tolerance. Record floor slope and surface condition. Exact rectangles, the eight variants and human scale proxies are in mockup.json; proxy figures are illustrations, not an anthropometric sample.

Materials and assembly

Borrow six undamaged, appropriately rated commercial chairs where available. The current Lifetime 2802G manufacturer specification gives an overall 21.4 in depth, 19.2 in width and 33.8 in height; manufacturer answers state 500 lb capacity and a 16 × 17 in seat. Verify the actual model, rating, setup, condition and stable floor contact before use. Approximately 17.5 in seat height differs from the 18 in design target: record it, without shimming. [C3EM06-07]

Two such chairs occupy 38.4 in overall width. They fit the baseline 40 in bench envelope, leaving only 1.6 in total. They cannot represent a pair within the 38 or 36 in alternative bench widths. Their individual contoured seats also cannot establish comfort on a continuous 40 in two-person bench. A suitable rated bench or professionally released seating prototype is needed for that decision. Record actual chair seat-front offsets; the 21.4 in footprint exceeds the 20 in schematic bench envelope by 1.4 in. At 32 in pitch, the idealized overall-chair gap is 10.6 in, not the 12 in bench-envelope gap. Actual feet and occupied clearances still require measurement.

Use four 2 in × 60 yd removable-tape rolls as a base allowance, two lightweight 48 × 96 in sheets, a 50 ft measuring tape, straightedge, square, labels and clipboards. Retaping every full-coach alternative may need up to four additional rolls; changing only the active bay reduces waste. Check tape compatibility with the existing floor. No quantity is a supplier quote.

Cut only visual templates, using the following sheet coordinates from each sheet's lower-left corner:

Sheet	Pieces	Width × length	Origins (x,y), inches
A	Four jamb strips	3 × 78	(0,0), (4,0), (8,0), (12,0)
A	Two bench plan templates	20 × 40	(17,0), (17,42)
B	Five bench back cards	40 × 16	(0,0), (0,18), (0,36), (0,54), (0,72)

These pieces fit with positive cutting gaps. Round exposed card corners. Trace the plans and remove loose templates before occupied movement. Hold jamb strips only during unoccupied observation or a stationary photo; install no overhead header. Do not use cards as seat pans, back supports, steps or barriers.

Mark the complete regular layout, then place six chairs in the active two-row area. The schematic seat reference centres are (105,−42), (105,−22), (105,22), (105,42), (137,−42), (137,−22). Align actual seat fronts and record offsets. Photograph the empty layout with a scale and case ID before participants enter. Keep the real room exits open throughout.

Six-adult sequence and decisions

Use anonymous P1-P6 with varied adult sizes and winter clothing where practical. Participation and photography are voluntary. Record measured dimensions rather than assigning the drawing figures to real people.

1. Six adults sit for ten minutes at the 32 in pitch. Record knee clearance, lateral intrusion and a comfort score from 1 (unacceptable) to 5 (comfortable). Any forced posture, contact, unusable position, score 1-2 or request to stop requires revision. Higher scores are sample evidence, not capacity approval.

2. Retain four seated adults. The other two walk single file each way at their chosen pace, then exchange roles. Record contact, foot catches, minimum dynamic clearances and backtracking. Do not force passing or infer standing passenger capacity.

3. Ask a window-side participant to leave the row toward each end. Record whether the aisle occupant must stand and how many movements are required. Empty taped rows do not reproduce a fully occupied coach.

4. Change one variable per labeled case. For 24/28/32 in aisles, each bench becomes 40/38/36 in: two nominal places receive 20/19/18 in each. Narrower paired-seat cases stay unoccupied with these chairs. For 32/34/36 in row pitches, the rear end zone falls from **31 to 17 to 3 in**. More knee room can make the rear exit unusable.

5. Compare 32 and 36 in regular end-door widths and record reach, turn and approach needs. Observe head height without an overhead prop. The modeled 78 in opening remains a review item; a tall adult fitting through it is not code evidence.

Use the observation, participant, chair and device forms. Record case, actual dimensions, device ID, contacts, assistance requested, time, photos, reviewer and retain/revise/reject decision. Missing observations remain blank, never zero or a pass. No evacuation-time threshold is invented here.

Accessible boarding and evacuation evaluation

Retape the accessible arrangement. Begin with empty representative devices, including the intended larger powered-device envelope, then use voluntary participants controlling their own devices. Record overall dimensions, footrest sweep, turning method and attendant position. Enter through both side doors, turn into each bay, test the two-bay arrangement, companions and the route to each end. Do not force a maneuver or carry anyone. One device succeeding does not prove all-device access.

The Access Board distinguishes amusement rides from vehicles primarily used for transportation. North Pole disembarking makes the actual service description relevant, but does not settle classification. The responsible accessibility professional and authorities must resolve the applicable framework. Private, seasonal operation alone does not establish the mobile-ride exception. [C3EM01, C3EM03]

Where the amusement provisions apply, side-entry maneuvering and companion relationships need more than nominal rectangles. The 42 in opening and 60 × 36 in bay are advisory starting points; companions across the route are not automatically shoulder-to-shoulder compliance. Boarding gaps and surfaces require loaded interface review. The bridge exception refers to specified transportation-vehicle provisions, not automatic application of every light-rail rule. The 24 in regular aisle and 36 in accessible route remain unapproved. [C3EM02, C3EM04-05]

Rehearse normal level-platform boarding using the flat 36 × 36 in bridge outlines and 60 × 60 in station spaces. Observe alignment, operator access, proposed gates and sequence at both stations. Actual bridge bearing, retention, surface, load rating and loaded movement require separate engineering. A 36 in horizontal bridge run would accommodate only 3 in rise at a 1:12 planning slope; actual platform levels are not fixed.

An optional lift reservation is x = 4.52-55.48, y = 34-52. A separate 48 in door trial is x = 6-54; moving the turning-circle centre to (30,3) creates only 1 in mathematical separation. The BraunAbility reference is not a rail-approved installation or exact stowed envelope. Keep this trial unoccupied until OEM CAD, loads and specialist installation are resolved. [C3EM08]

For emergency planning, an observer declares one planned exit unavailable without blocking a real exit. Compare: recovery to a safe station using an engineered operational procedure; an independent rated lift or deployable accessible route at a suitable wayside; and a trained assisted-transfer arrangement for users who choose or require it. Record who brings equipment, where it fits, communications, lighting and device recovery. These are alternatives to evaluate, not an accepted rescue plan. No blanket manual-carry assumption is allowed; elevated transfers need suitable equipment and trained personnel.

Floor-height consequences and release boundary

All occupied trials remain on the slab. E0/E2 retain the 28 in floor; E1 studies 32 in. Using a **planning sensitivity** of maximum 7 in rise and 11 in going, rail-level boarding would require four 7 in risers and 33 in run at 28 in; five 6.4 in risers and 44 in run at 32 in; or six 6 in risers and 55 in run at 36 in. A 1:12 ground ramp requires respectively 28, 32 or 36 ft horizontal run, plus landings. These are diagrammatic consequences, not released stair geometry.

The F075 floor stack remains unchanged. E2's locally rotated members have greater calculated support movement, so panel behavior alone cannot qualify the floor. This mockup neither proof-loads that structure nor clears wheel, suspension, stair or coupling envelopes. Release seating, doors, companion arrangement and boarding interfaces only after measured evidence and specialist review; preserve the unperformed and failed cases for the next design revision.

06 | Supplier evidence and first-article verification

PF-TRAIN-S3E-R0 · 18 October 2026 UTC · **Prepared technical inquiries only: unsent, unquoted and unapproved. Test records are blank; no test has been performed.**

Turn the studies into comparable, reviewable offers

The next purchase decision should be based on a configured system and accepted interfaces. Seven inquiry packets now define what each supplier must return, which drawings it must address, who reviews the evidence and what depends on it. These packets consolidate the earlier open component fields into an offer-comparison and acceptance process. They do not authorize outreach or orders.

Every packet retains one diesel locomotive, three regular coaches and one accessible coach; 36-inch gauge evaluation; a 0% longitudinal grade target; 13,200/11,000 kg upper coach analysis cases and 65,100 kg whole-train upper case. The 7,040/7,535 kg tare allocations remain unchanged. Speeds of 5/7.5 mph and radii of 30/50/75 m are sensitivities, not operating approvals. A supplier must explicitly accept, qualify or reject these cases rather than replacing them with an advertised passenger count.

The Phase 3E floor/frame alternatives must receive separate responses: **E0** retains the 28-inch floor and D2 arrangement; **E1** studies a 32-inch floor; **E2** retains 28 inches while changing local members and inner seat rails. None is selected. With F075, the central-bolster underside is 381.1 mm for E0/E2 and 482.7 mm for E1. Actual support height, attachment stack and relative motion remain unknown. A supplier's static dimension is insufficient to close a moving-clearance problem.

Seven coordinated technical packages

Packet	Initial scope and required decision	Review lead
P01 - Complete bogies	Two braked bogies for the first coach; identify design authority, accepted load cases, wheel/axle sharing, support system, moving CAD and actual mass/CG	Vehicle engineer, bogie OEM and track engineer
P02 - Wheelsets	Controlled wheel profile, axle/bearing assembly and matching rail/check-gauge/flangeway record through the bogie design lead	Track and vehicle engineers with wheelset supplier
P03 - Connections	Four vehicle joints, distinguishing locomotive-operating and semi-permanent coach interfaces; configured ratings, articulation, retention, stroke and frame reactions	Structural engineer, coupler OEM and brake integrator
P04 - Brakes	Matched first-coach package plus locomotive/full-train controls; service, emergency, parking and separation behavior	Brake design authority and vehicle/locomotive engineers
P05 - Floor	Directional material evidence, representative support/seam assembly and separate seat/wheelchair/lift hard points	Body/structural engineer, panel OEM and laboratory
P06 - HVAC	Regular and accessible configured packages, delivered thermal/ventilation duty, input/start data, mounts and service envelopes	HVAC/body engineers and electrical integrator
P07 - Hotel power	Site-rated generation, shore transfer, coach distribution, independent brake/lighting power and protection	Electrical engineer, generator OEM and brake integrator

The packets contain exact return lists and links to current Phase 3E drawings and retained Phase 3D reference sheets. P01/P02 must close together: nominal gauge does not settle tread/flange contact, back-to-back, rail inclination or crossing geometry. The wheelset specialist must work under the bogie design lead so separate purchases do not leave an unowned compatibility problem.

P03 must distinguish working, fatigue, proof and ultimate ratings. Neither a catalog family strength nor the retained quasistatic draw force establishes attachment design. Its return must cover combined longitudinal/lateral/vertical forces, slack impact, reversing, wear, guards and cable separation. The existing crossmember in the concealed connection route is a packaging issue to resolve in the actual assembly, not an invitation to cut a clearance hole.

P04 asks for one accountable brake design authority. Separate Knorr-Bremse and Hanning & Kahl catalog components are not a verified assembly. The return must include force convention, force-pressure-travel maps, fluid displacement, electrical demand, software/configuration control and the complete emergency application path. A second hydraulic release circuit releases a spring brake; it does not establish independent emergency application. The documented EHM is a product reference, not the chosen circuit. [C3EV06]

P05 asks Plascore to reconcile its published rigidity discrepancy and identify production minima, orientation and separated bending/shear properties. Its build-to-specification offering supports a defined prototype inquiry. [C3EV04] Public ASTM C393 scope identifies relevant sandwich characterization, but the laboratory must obtain and adopt the applicable complete procedures; this report supplies no standard test load or pass criterion. [C3EV03]

P06/P07 must resolve voltage and thermal duty together. The retained cooling scenario must be recalculated from the actual envelope; the proposed insulation does not substantiate the earlier opaque-U assumption. The 101.2 kW heating example remains a resistance-only sensitivity, not verified electrical demand. Northwest Rail Electric's package configurations vary with compressor selection. [C3EV07] Stadco's reference ratings and starting-kVA figures require site derating and actual transient acceptance. [C3EV05]

Review evidence without confusing receipt with acceptance

The new matrix has four decisions per package: configuration/responsibility, interfaces, performance envelope and inspection/support/delivery basis. Each row records the evidence ID, controlled revision, reviewer, decision and related deviation. All actual fields are blank. “Received” means only that a file arrived; “accepted” requires the named reviewer and the exact configuration to which the decision applies.

Use the deviation form for every changed component, floor height, wheel, seat rail, software revision or supplier exclusion. Record affected drawings, load paths, mass/CG, utilities, tests and maintenance. The engineer defines the required recalculation or regression tests and signs the disposition. A substitute bearing or brake valve cannot inherit acceptance merely because it has similar catalog dimensions.

Keep raw supplier documents, revised drawings, calculation references and test files under unique IDs. A superseded document remains traceable. The portable generator preserves any differing existing template in a content-hashed archive before regeneration; completed working records should use separate run/configuration filenames. Commercial terms can be retained separately from the public technical portal. Generic certificates and marketing claims cannot fill configuration-specific evidence fields, and no checklist automatically provides a construction release.

Procurement sequence with explicit gates

G00-G20: name the responsible design team, prepare authorized inquiries, compare exact configurations and close coupled interfaces. Do not freeze a floor height independently of bogies, stairs, accessible boarding, body height and CG. The actual station and track interface remain engineer-coordinated; deferring the North Pole location does not prevent vehicle interface review.

G30-G50: release controlled prototype drawings, guarded laboratory methods and inspection plans; obtain owner authorization for any order; inspect the first article; complete bench and secured-coach evidence before limited running. Prototype laboratory specimens can be released separately when their scope and test methods are defined. A bare panel coupon cannot qualify the complete assembled floor.

G60: after first-coach dynamic acceptance, release the remaining three coaches only to the accepted configuration and within owner purchasing authorization. The accessible coach requires its own load paths, boarding interfaces and qualification. One regular coach's successful test cannot approve those differences.

G65-G80: assemble and inspect the exact full consist, including every vehicle and joint; complete the full-train unoccupied campaign and site commissioning; close operator training, maintenance and evacuation evidence before passenger-service release. Later fleet production must preserve the accepted configuration or return to affected gates. This sequence avoids requiring full-train acceptance before the remaining cars can be built, while preventing first-coach results from being treated as train-level proof.

There are no invented delivery weeks. Each supplier receives a dated milestone RFI covering engineering data, interface review, drawing release, material availability, factory slot, build duration after release, FAT booking, retest allowance, shipping/import, installation and commissioning support. Require dependencies and validity for each answer. The critical path can be scheduled when these returns exist; a website's generic delivery statement is not a commitment.

Instrumentation and records for the first article

Seventeen measurement functions define the data needed before instruments are purchased. They include common command/fault timestamps, independent ground speed/distance, four axle speeds, both working and auxiliary hydraulic zones, actuator travel or an accepted force proxy, body acceleration, brake temperature, connection force/stroke, brake DC, hotel AC, HVAC conditions, emergency lighting and floor/seam displacement, calibrated wheel loads and independent support movement.

The first-coach reference stations are x = 45 and 261 inches at the bogies and x = 153 inches at the body centre. These locate measurement functions, not mounting holes or pressure fittings. Instrumented couplers and hydraulic connections require OEM/engineer-designed arrangements that preserve retention and circuit integrity. Do not insert an ordinary load cell into an unreviewed drawbar.

For the floor study, two full-width panels occupy x = 93-141 and 141-189 inches, with supports at 93, 117, 141, 165 and 189 inches and a seam at 141. A trial load-patch centre at x = 129, y = 0 identifies one

mid-bay measurement location. Patch dimensions, loads, repetitions, closures, attachments and acceptance values remain blank for the laboratory protocol. Additional seam, seat and wheelchair cases must use actual contact and anchor geometry. Measure panel movement against an independent stable datum while recording support movement simultaneously. Report panel-relative deflection and total floor movement separately; the E2 frame cannot be assumed rigid.

A nominal 100 samples/s synchronized vehicle-data stream and no more than 10 ms timing mismatch are **instrument-proposal targets**, subject to the engineer's bandwidth and uncertainty review. At 7.5 mph, 10 ms corresponds to about 34 mm of travel; timing error can materially affect response-distance comparisons. Electrical starting waveforms need a suitable power analyzer and separate acquisition settings. Do not treat the nominal vehicle stream as adequate for every electrical transient.

Record sensor serial number, range, calibration, uncertainty, orientation, sample rate, filtering and clock synchronization. NIST TN1297 supports explicit uncertainty evaluation and reporting, but provides no train stopping acceptance limit. [C3EV01] Use an independent ground reference: a locked axle can indicate zero wheel speed while the car is moving. Preserve raw data and declare any filtering used to calculate jerk or detect force onset.

A staged test campaign

The fourteen test definitions cover receiving/wheel-load baseline; floor/seam qualification; guarded actuator/hydraulic maps; secured service/emergency/reset; secured separation/data faults; parking/recovery; single-coach running; full-consist brake balance; adhesion/environment; electrical source/protection; HVAC duty/start; blackout/E-stop/lighting; installed boarding-equipment qualification; and operational handover.

Begin with OEM or qualified-laboratory fixtures. Hanning & Kahl documents dedicated hydraulic test equipment, reinforcing the need for a defined specialist setup rather than an improvised live-pressure trial. [C3EV02] Pressure settings, proof forces, maximum temperatures and failure injection must be specified in the approved protocol. Farm personnel must not cut a live hose, defeat guards or place people in proof-loaded structures.

On a secured unoccupied coach, distinguish loss of hotel AC, loss of brake DC, service-bus failure, emergency-circuit faults, blocked release paths and auxiliary recovery states. Verify traction inhibition, local application, deliberate reset and restoration. Initially simulate mechanical separation while stationary, including the case where electrical jumpers remain attached. A moving breakaway experiment requires its own engineered containment and recovery plan.

Running trials follow an approved progression from low speed and known loading to the authorized target cases. Record actual mass, wheel reactions, direction, initial speed, temperature, rail condition and configuration for every run. Use controlled ballast and restraints, not passengers as test weights. Full-consist trials measure each vehicle's response and connection loads; repeat appropriate cases in reverse travel. Adhesion trials require a specialist method and containment; casually contaminating rail is not an accepted test procedure.

Hot/cold and repeated-stop tests must address temperature-dependent response and fade. Parking criteria must include engineer-defined disturbances, residual-grade tolerances and duration even though the design grade is zero. Lighting reserve duration comes from the accepted evacuation plan. HVAC trials record ventilation, imposed thermal loads, door cycles, condensate and generator restart behavior, rather than simply confirming that air feels cold.

Practical responsibilities and closeout

Before any supervised participant exercise on installed vehicle/boarding equipment, E14 requires accepted installed bridge/lift and evacuation-equipment evidence at both stations. Engineer/OEM protocols must address actual loads, deployment, retention, stowage, movement inhibits, power-loss recovery and alternate or wayside equipment. Floor-tape mockups cannot close this gate. E13 then uses an approved supervised procedure to examine normal boarding, blocked exits and recovery without exposing people to unqualified equipment or unsafe fault injection.

Farm crew can advance nonstructural mockups, inventory photographs, labeled records and approved survey/track preparation. After specific release, they can assist with body finishing, routing supports, barriers and measurements. Wheelsets, critical frame/anchor welding, brake plumbing/controls, energized electrical work, proof fixtures and acceptance interpretation remain specialist responsibilities.

Every run needs a test director, approved method, limits, calibration check, containment/restoration plan and witnesses. An unexpected motion, failed restraint, loss of critical measurement, leakage or result outside an approved limit stops that test under its protocol. Preserve failed and aborted records; criteria cannot be revised after a run to manufacture a pass. The final evidence bundle links the delivered serials and software, accepted drawings, raw measurements, deviations, training, inspection intervals and explicit release decisions.

07 | Files, responsibilities and limits

Deliverable	Use and limitation
PDF / SVG	Review narrative and dimensioned sheets; all safety-critical unselected details remain explicitly held.
Full-size DXF	Coordinate layouts in inches for mockup marking and alternative frame review. Paper-sheet DXFs are separately identified; never scale a printed sheet to infer an omitted dimension.
STEP / rotating models	Millimetre STEP solids and inch browser meshes show nominal members, floor-panel envelopes and identified wheel proxies. They omit manufacturing joints, actual wheel profiles and a configured bogie.
XLSX / JSON / scripts	Traceable sensitivity calculations and blank actual observations. Formula checks are separate from physical validation and engineering acceptance.
Supplier and test forms	Unsent technical packets and unperformed test records. Price, lead time, availability, actual ratings and signoffs must be supplied by the responsible parties.

Farm-crew work in this issue is the level-ground mockup, organized measurements and the information collection assigned in the responsibility matrix. Specialist work includes configured running gear, structural joints and welding procedures, brake/control integration, electrical protection, accessibility engineering, survey/track acceptance and controlled vehicle tests. The complete car remains a coordinated engineering responsibility.

The verification record distinguishes geometric/export checks, independent calculations, document rendering and live browser interaction. No test result or professional approval is implied by successful file generation or site publication.

08 | Source register

Primary publications and retained project calculations are listed with their access dates and limits. Publication availability does not establish stock, lead time, quoted price or project-specific acceptance.

C3ES01 | Atlas Tube ASTM A500 rectangular HSS section properties

Accessed 2026-10-18

4x2x1/8 catalog strong I2.65/S1.33, weak I0.898/S0.898, nominal4.75lb/ft, designwall0.116in. Existing12x4x1/4 and3x1x1/8 sections retained from manufacturer table.

Limit: Current manufacturer-hosted table access is not stock, material certification, allowable stress, weld/fatigue or rail application approval.

https://www.atlastube.com/wp-content/uploads/2018/04/Atlas-HSS-Section-Properties-A500-Rectangles.pdf

C3ES02 | Steel Tube Institute HSS combined shear, axial and in-plane moment example

Accessed 2026-10-18

HSS member and local joint behavior require combined-force and wall checks; no project weld dimensions adopted.

Limit: General technical example, not the vehicle load standard or a released connection.

https://steeltubeinstitute.org/resources/hss-to-hss-connection-with-shear-axial-and-in-plane-moment-example/

C3ES03 | Retained Phase3D structural and running-gear study

Accessed 2026-10-18

D2 110member1527.287845kg benchmark,24in nodes,12inmains/bolsters and nominal596mm wheel/crossmember conflict retained as comparison inputs.

Limit: Preliminary source models, not physical measurements or OEM approval.

C3ES04 | Phase3E coordinated design basis

Accessed 2026-10-18

0%grade,28in baselinefloor,32regularplaces,13200kg upper structural case and7040kg tare allocation retained.

Limit: Owner/analyst design basis; alternative32in floor is not selected.

C3EC01 | Railteco corporate passenger-bogie catalog

Accessed 2026-10-18

RTKZ1 forest-passenger bogie described as762mm gauge with596mm wheels, centreplate and sprung sidebearers.

Limit: Different gauge; no project support, profile or suspension envelope. Source translated from Chinese; product not selected.

https://cn.railteco.com/cplist.aspx?cptp=8&i=14

C3EC02 | Railteco RTKZ1 manufacturer sales technical table

Accessed 2026-10-18

596mm diameter and1165mm wheelbase; published3.75t maximum axle and<=1.5t bogie tare.

Limit: Retain3D rejection as supplied;762m/RTKZ32 copyerrors require controlled document. These two dimensions only anchor a proxy; no offered914.4mm design or profile is inferred.

https://www.railwayspare-parts.com/sale-13771970-aar-standard-762m-gauge-passenger-train-bogie.html

C3EC03 | GHH-BONATRANS custom wheelsets

Accessed 2026-10-18

Manufacturer develops wheelsets around customer vehicle, bogie, brakes, drive and track requirements.

Limit: No selected part, profile,load or dimensions. Supports coordinated engineering route only.

https://www.ghh-radsatz.com/en/products-and-services/wheelset/

C3EC04 | Retained Phase3D running-gear calculation and source record

Accessed 2026-10-18

Local predecessor preserved;11.7/37.1mm nominal conflict and110member D2 scheme carried forward.

Limit: Project analytical evidence, not external approval;3E extends checks to every member and explicitly bounded motion proxies.

C3EM01 | US Access Board - ADA Chapter 2, §234

Accessed 2026-10-18

New amusement ride scoping addresses load/unload areas and wheelchair or transfer alternatives. The mobile/portable exception concerns regularly assembled and disassembled rides.

Limit: Seasonal operation on private land does not establish this exception. Project classification and complete applicability need responsible review.

https://www.access-board.gov/ada/chapter/ch02/

C3EM02 | US Access Board - ADA Chapter 10, §1002

Accessed 2026-10-18

Addresses wheelchair entry, companions and boarding interfaces. Side entry advisory associates a 42 in opening with a 60 × 36 in maneuvering space.

Limit: Advisory dimensions do not validate this layout or every device. Companion position, routes, gaps, slopes and operations need full review.

https://www.access-board.gov/ada/chapter/ch10/

C3EM03 | US Access Board - Amusement Rides guide

Accessed 2026-10-18

Distinguishes amusement systems from vehicles primarily transporting people, which are addressed by transportation vehicle guidelines.

Limit: Train name, gauge and North Pole disembarking do not determine the classification. Jurisdiction remains unresolved.

https://www.access-board.gov/ada/guides/chapter-10-amusement-rides/

C3EM04 | US Access Board - Light Rail, §1192.83(c)

Accessed 2026-10-18

The bridge/ramp reference covers load, surface, width, thresholds, edges, slope, retention and stowage. The amusement gap exception references applicable portions.

Limit: This cross-reference does not apply the entire Light Rail rule to this train. Minimum prescribed loads are not necessarily adequate project loads.

https://www.access-board.gov/ada/vehicles/subpart-d/

C3EM05 | US Access Board - ADA Chapter 4

Accessed 2026-10-18

Accessible route and door maneuvering requirements apply where scoped; nominal aisle width alone does not establish compliance.

Limit: Do not infer compliance of the 24 or 36 in vehicle aisles from one building dimension.

https://www.access-board.gov/ada/chapter/ch04/

C3EM06 | Lifetime - Commercial folding chair 2802G

Accessed 2026-10-18

Current specification: overall 21.4 × 19.2 × 33.8 in. Manufacturer Q&A reports 500 lb capacity and 16 × 17 in seat.

Limit: Verify the actual model, label, instructions, condition and dimensions. Individual chairs do not validate a continuous two-person bench.

https://www.lifetime.com/lifetime-2802G-classic-folding-chair-commercial

C3EM07 | Lifetime - 2802G manufacturer Q&A, page 2

Accessed 2026-10-18

Manufacturer answers repeat 500 lb capacity and approximately 17.5 in seat height; older answers give 19.1 in overall width.

Limit: Use the current 19.2 in specification for screening and measure the actual unit. The earlier catalog 18 in width is superseded for this mockup.

https://www.lifetime.com/lifetime-2802g-classic-folding-chair-commercial?bvstate=pg%3A2%2Fct%3Aq

C3EM08 | BraunAbility - NCL1000 application guide, 3 June 2026

Accessed 2026-10-18

Retained optional NCL1000IB3754HB-2 reference: 50.96 in width, 45.85 in minimum door, 18 in threshold reservation and 1000 lb rating.

Limit: Exact stowed CAD, mounts, rail application, device/load fit and rescue remain unresolved. No occupied mockup lift deployment.

https://www.braunability.com/content/dam/braunability/docs/comm/na/app-guide/updated-dpa-lift-app-guides/Lifts-Comm-Century-1000-Guide-6-3-26.pdf

C3EV01 | Technical Note 1297: evaluating and expressing measurement uncertainty

Accessed 2026-10-18

Current public page presents the 1994 edition; measurement uncertainty components, combination and reporting guidance.

Limit: General metrology guidance, not train stopping criteria.

https://www.nist.gov/pml/nist-technical-note-1297

C3EV02 | Hydraulic test bench with VDT

Accessed 2026-10-18

OEM offers purpose-built hydraulic brake test/service equipment.

Limit: No project fixture design, pressure or test limit inferred.

https://www.hanning-kahl.com/service/testing-and-service-equipment/testing-and-service-equipment-brakes/hydraulic-test-bench-with-vdt.html

C3EV03 | C393/C393M-20 public scope

Accessed 2026-10-18

Public page identifies sandwich beam core/shear and load-deflection characterization; distinguishes properties requiring additional methods.

Limit: Full standard not obtained; laboratory must select licensed edition/procedure. No test method or acceptance value reproduced.

https://store.astm.org/c0393_c0393m-20.html

C3EV04 | Base panels

Accessed 2026-10-18

Offers build-to-print/build-to-specification panels; published typical properties do not settle project qualification.

Limit: Actual construction, directional/design data and service acceptance held.

https://plascore.com/composite-panels/base-panels/

C3EV05 | Mark 6 S32.25

Accessed 2026-10-18

Reference continuous output and motor-start ratings require attention to stated conditions and climatic derating.

Limit: Catalog is not guaranteed installed site performance.

https://stadcogen.com/wp-content/uploads/2025/05/S32.25-Mark-6-Specs.pdf

C3EV06 | EHM and system configuration

Accessed 2026-10-18

Documented two-bogie electro-hydraulic reference with specific power/control voltage and permissible pressure.

Limit: No cross-manufacturer application acceptance or selected pressure.

https://www.hanning-kahl.com/products/hydraulic-brake-systems/electro-hydraulic-module.html

C3EV07 | NW-64-CN customized rail condenser

Accessed 2026-10-18

Compressor/package configurations vary; matched configuration must be identified.

Limit: Nominal tons do not establish coach capacity or electrical starting demand.

https://www.nwrail.com/nwre_products/64cn.html